Shiv Shankar Singh

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Summary

Pre-final-year undergraduate student of Electrical Engineering at NIT Jalandhar with a strong interest in **Artificial Intelligence** and emerging technologies. Self-motivated learner and problem solver who enjoys exploring new concepts, tackling complex problems, and implementing practical solutions. Values continuous learning, adaptability, analytical thinking, and teamwork to drive effective problem solving.

Education

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar **Expected 2028**
B.Tech in Electrical Engineering

Saint Fateh Singh Convent School, Bathinda, Punjab
Schooling (PCM)

Projects

Fiat | *Multi-Agent AI Coding Agent* | GitHub | PyPI *Python, LLMs, Multi-Agent Systems, CLI*

- Architected a **provider-agnostic, terminal-native coding agent** in which **Planner, Coder, Tester, and Reviewer** agents collaborate to plan, implement, test, and review code iteratively.
- Engineered the orchestration layer with **shared runtime state**, configurable agent iterations, automated test/review feedback loops, and **role-based tool permissions** to constrain code execution.
- Built a provider abstraction over **Gemini, OpenAI, Anthropic, OpenRouter, and Groq** with runtime model switching and streaming; shipped an interactive CLI with slash commands, context/token/cost monitoring, and secure credential management, published on **PyPI**.

SAGE | *Autonomous Research & Presentation Studio* | GitHub | Live Demo *React, TypeScript, FastAPI, LangGraph, Supabase*

- Built and deployed a full-stack multi-agent studio that turns one topic into a cited Markdown article and a slide deck through a **5-stage LangGraph pipeline** (Router, Research, Planner, parallel Workers, Reducer), with live web research via **Tavily** and source deduplication.
- Parallelized section drafting with LangGraph's **Send API** under concurrency limits (**3 workers** on Groq, **5** on frontier providers) and exponential backoff on rate limits.
- Generated lecture-grade decks across **9 slide layouts**, compiled to both a **Beamer PDF** (Tectonic) and an editable **16:9 PPTX** (python-pptx), with a deterministic linter and a single-slide LLM repair loop.
- Built a bring-your-own-key layer across **5 LLM providers** with Fernet-encrypted keys, **Supabase** auth (email and Google OAuth) with row-level security, and rate limits (**20** guest jobs/hour; **3** concurrent, **50** daily for signed-in users); deployed with Docker on Render and Vercel.

Experience

Research Intern **Oct 2025 – Mar 2026**
OpenLearn, NIT Jalandhar *Offer Letter | Letter Of Recommendation*

- Reviewed and compared existing genomics-focused deep learning models, analyzing their strengths and limitations to identify directions for improving them.

Achievements

- **Top 10** — Makeathon, Thapar Institute of Engineering & Technology (National-Level Hackathon)
- **Top 250** — Amazon AI for Bharat (10,000+ participants)
- **300+ DSA Problems** solved on LeetCode

Skills

Programming: Python, C++, JavaScript, TypeScript, SQL

Software Engineering: Software Engineering, Web Development, Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Git, GitHub

Frameworks & Tools: React, FastAPI, Tailwind CSS, Docker, LangGraph, LangChain, Pydantic, Supabase, Jupyter, Vercel, Render

AI/ML & Data: PyTorch, Scikit-Learn, Hugging Face, Deep Learning, Transformers, LLMs, RAG, Agentic AI, Data Analysis

Soft Skills: Problem Solving, Analytical Skills, Communication, Collaboration